K-FLATNESS MAKES THE COMPOSITION OF DERIVED TENSOR FUNCTORS COMPUTED BY PLAIN TENSORS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: dependencies carried as navigation stubs: the property (P) resolution construction behind equation equation-plain-versus-derived-algebras and Lemma lemma-compose-tensor-functors-general-algebra (both in-source), plus the standard More on Algebra K-flat quasi-isomorphism lemma (external tooling).

Editorial note. The label lemma-compose-tensor-functors names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Property (P) resolutions compute derived tensor. Same unit, the construction behind equation equation-plain-versus-derived-algebras and Lemma lemma-compose-tensor-functors-general-algebra. The proof chooses a property (P) resolution P -> (A tensor_R B)_B and works with the plain tensor representatives of the derived functors as in that construction.
- K-flat base change. More on Algebra, Lemma more-algebra-lemma-K-flat-quasi-isomorphism (standard tooling). Final step (source line 6090): tensoring a quasi-isomorphism with a K-flat complex of R-modules preserves quasi-isomorphism, so P tensor_R C -> A tensor_R B tensor_R C is a quasi-isomorphism when C is K-flat.

Source reference. Original equation 0BZ4: sources/stacks-dga/pinned-original-source.tex, source lines 5941–5941. Prerequisite pointer invoked at proof line(s) 6071; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BZ5: sources/stacks-dga/pinned-original-source.tex, source lines 5959–5959. Prerequisite pointer invoked at proof line(s) 6073; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 06Y0: sources/stacks-chow/tags.txt, source lines 8869–8869. Prerequisite pointer invoked at proof line(s) 6090; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:dga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let R be a ring. Let (A, d), (B, d), and (C, d) be differential graded R-algebras. If C is K-flat as a complex of R-modules, then (0BZ4) is an isomorphism and the conclusion of Lemma 0BZ5 is valid.*

Proof. Choose a quasi-isomorphism $P \to (A \otimes_R B)_B$ of differential graded B-modules, where P has property (P). Then we have to show that

$$P \otimes_B (B \otimes_R C) \longrightarrow (A \otimes_R B) \otimes_B (B \otimes_R C)$$

is a quasi-isomorphism. Equivalently we are looking at

$$P \otimes_R C \longrightarrow A \otimes_R B \otimes_R C$$

This is a quasi-isomorphism if C is K-flat as a complex of R-modules by More on Algebra, Lemma 06Y0. □